TCS NQT PREPARATION GUIDE

Data Interpretation

Complete Notes with 30+ PYQs | Charts & Tables

Bar Charts • Pie Charts • Line Graphs • Data Tables

■ Updated 2024-25	■ TCS NQT Pattern	■ 30+ PYQs Solved	■ 30-40 Min/Section

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1. Introduction to Data Interpretation

What is Data Interpretation?

Data Interpretation (DI) is the process of making sense of numerical data collected, analyzed and presented visually. In TCS NQT, DI questions test your ability to read, analyze and draw conclusions from data presented in charts, graphs and tables. It is one of the highest-weightage topics in the Numerical Ability section.

TCS NQT Exam Pattern – DI Section

Parameter	Details
Section	Numerical Ability
No. of DI Questions	10–15 (out of 26)
Time Allocated	~40 minutes
Difficulty Level	Moderate to High
Question Types	MCQ (4 options)
Negative Marking	No negative marking
Chart Types	Bar, Pie, Line, Table, Mixed

Types of Data Interpretation Questions

Type	Sub-types	Frequency in TCS NQT
Bar Charts	Simple, Double, Stacked, Horizontal	★★★★★
Pie Charts	Single pie, Multiple pies, Donut	★★★★■
Line Graphs	Single line, Multiple lines, Mixed	★★★★■
Data Tables	Simple, Cross-tabulation, Ratio	★★★★★
Caselet DI	Paragraph with embedded data	★★★■■
Mixed Graphs	Bar + Line combined	★★★■■

Essential Formulas

```
Percentage: Percentage = (Part / Whole) × 100
% Change: % Change = [(New − Old) / Old] × 100
Ratio: Ratio A:B = A/B → A = (A/(A+B)) × Total
Average: Average = Sum of all values / Count of values
Growth Rate: CAGR = [(Final/Initial)^(1/n) − 1] × 100
Fraction→%: 1/2=50%, 1/3≈33.3%, 1/4=25%, 1/5=20%, 1/8=12.5%
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■ TIP: In TCS NQT you get 2–2.5 min per DI question. Always scan ALL options before computing — elimination is faster than exact calculation.

2. Bar Charts – Theory & Examples

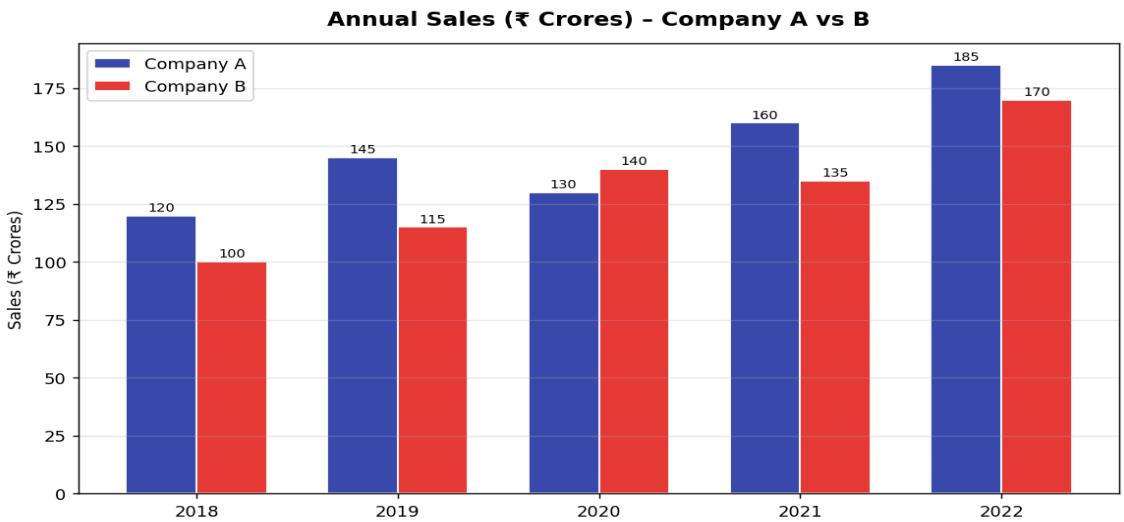
What is a Bar Chart?

A bar chart represents data using rectangular bars. The length/height of each bar is proportional to the value it represents. They can be vertical (column charts) or horizontal. Double/grouped bars allow comparison across two categories simultaneously.

Key Reading Strategy:

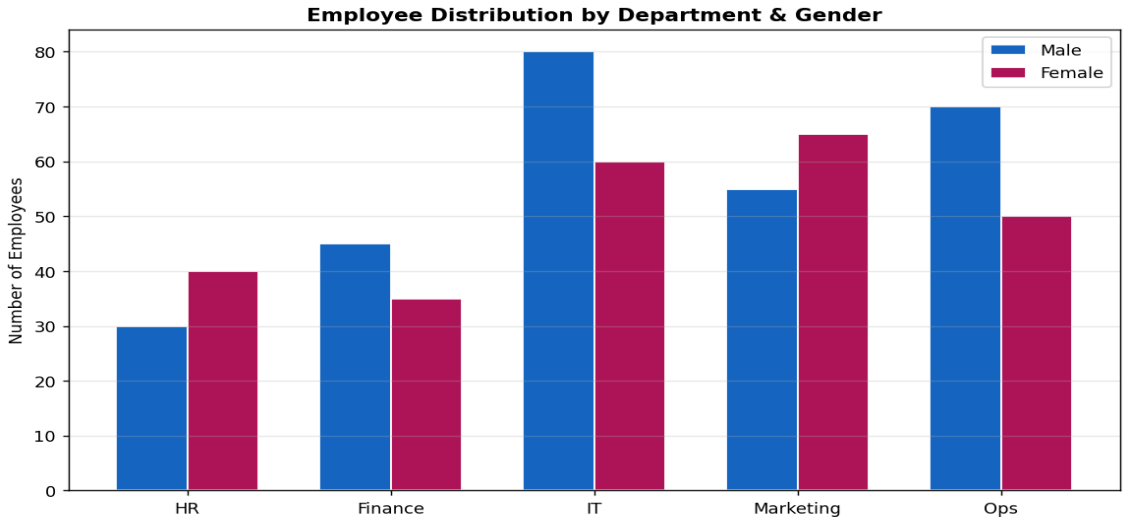
1. Read the title carefully — it tells you WHAT and WHEN.
2. Check both axes — note the unit (thousands, crores, %).
3. For double bars, note which color/pattern is which group.
4. Look for patterns: increasing, decreasing, fluctuating.
5. Estimate differences before calculating exactly.

Example 1: Annual Sales Comparison



Question	Solution	Answer
What is the % increase in sales of Company A from 2018 to 2022?	% increase = (185−120)/120 × 100 = 65/120 × 100 ≈ 54.2%	54.2%
In which year was the difference between A and B sales maximum?	2018: 20, 2019: 30, 2020: 10, 2021: 25, 2022: 15 → Max in 2019	2019 (■30 Cr)
What is the average sales of Company B over 5 years?	(100+115+140+135+170)/5 = 660/5 = 132	■132 Crores
Ratio of A's sales in 2020 to B's sales in 2021?	130 : 135 = 26 : 27	26:27

Example 2: Employee Distribution



Question	Solution	Answer
Total employees in IT department?	80 (M) + 60 (F) = 140	140
Dept with highest female employees?	Marketing: 65	Marketing
% of female employees in HR dept?	40/(30+40)×100 = 40/70×100 ≈ 57.1%	57.1%
Overall male to female ratio?	M=280, F=250 → 280:250 = 28:25	28:25

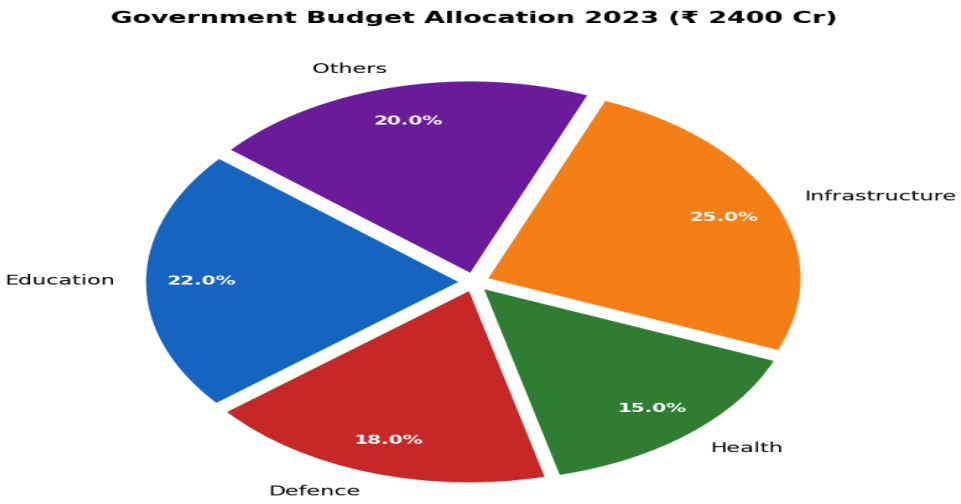
3. Pie Charts – Theory & Examples

Understanding Pie Charts

A pie chart (circle graph) shows proportional data as slices of a circle. The entire circle = 100% (or total). Each slice = one category's share. The central angle of a slice = (percentage/100) × 360°.

- **Value of segment = (% share / 100) × Total value**
- **Degrees of segment = (% share / 100) × 360°**
- **Ratio of two segments = Ratio of their percentage shares**
- **If total changes → new value = new % × new total / 100**

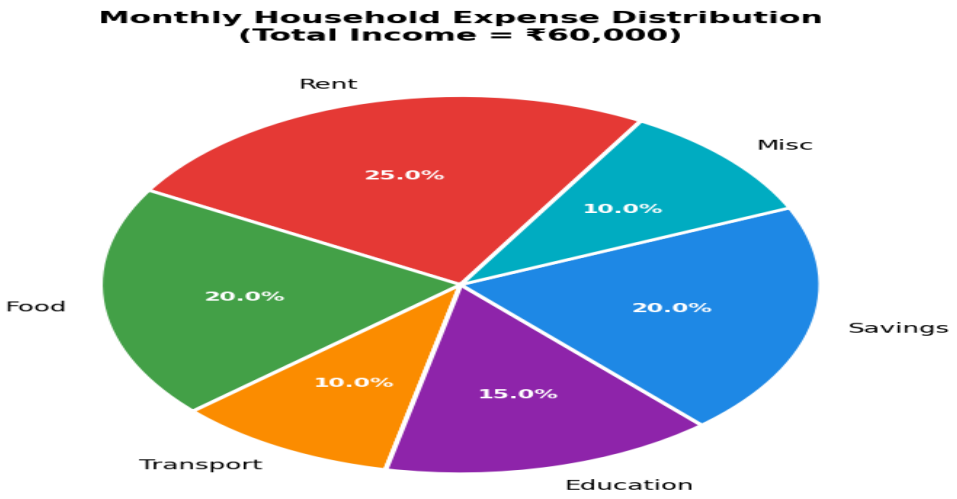
Example 3: Government Budget Allocation



Segment	% Share	Amount (■ Cr)	Degrees
Education	22%	528	79.2°
Defence	18%	432	64.8°
Health	15%	360	54.0°
Infrastructure	25%	600	90.0°
Others	20%	480	72.0°
TOTAL	100%	2400	360°

Question	Solution	Answer
How much more is spent on Infra vs Health?	600 – 360 = ■240 Crores	■240 Cr
What is the central angle for Education?	22/100 × 360 = 79.2°	79.2°
If Education budget grows by 10%, new amount?	528 × 1.10 = ■580.8 Crores	■580.8 Cr
Ratio of Defence to Others spending?	432:480 = 9:10	9:10

Example 4: Household Expense Distribution



Category	% Share	Monthly Amount (■)	Notes
Rent	25%	15,000	Largest expense
Food	20%	12,000	Necessary
Transport	10%	6,000	Variable
Education	15%	9,000	Fixed
Savings	20%	12,000	Investment
Miscellaneous	10%	6,000	Variable

4. Line Graphs – Theory & Examples

Understanding Line Graphs

A line graph shows trends over continuous time. Points are plotted and connected. Multiple lines on the same graph allow trend comparison. Line graphs are ideal for showing growth, decline, or fluctuation patterns over time.

Key Analysis Points:

• Upward slope → Increasing trend

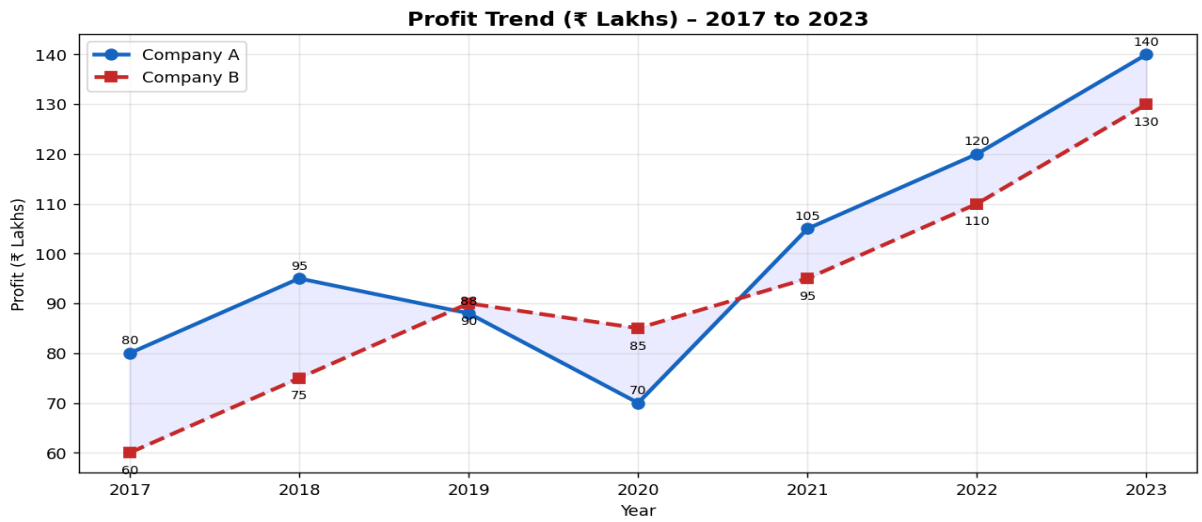
• Downward slope → Decreasing trend

• Flat/horizontal → No change

• Steeper slope → Faster rate of change

• Crossing lines → Values become equal at that point

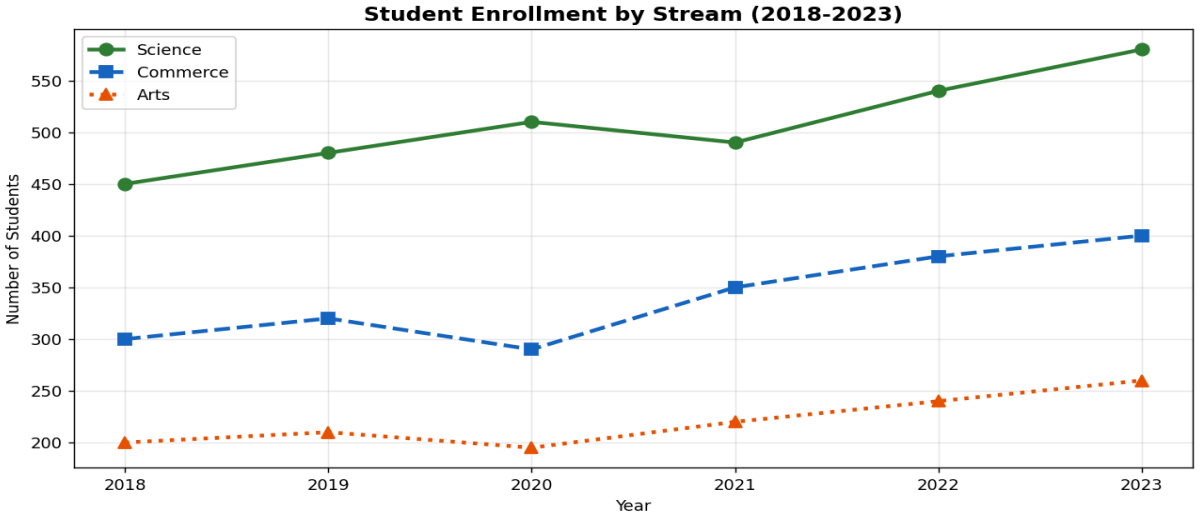
Example 5: Company Profit Trend (2017–2023)



Year	2017	2018	2019	2020	2021	2022	2023
Company A (■L)	80	95	88	70	105	120	140
Company B (■L)	60	75	90	85	95	110	130
Difference (A-B)	20	20	-2	-15	10	10	10

Question	Solution	Answer
In which year did Company B's profit exceed A's?	2019: B=90 > A=88; 2020: B=85 > A=70	2019 & 2020
% growth of A from 2020 to 2023?	(140–70)/70 × 100 = 100%	100%
Average profit of B from 2017 to 2023?	(60+75+90+85+95+110+130)/7 = 645/7 ≈ 92.1	≈ ■92.1 L
Year with highest % growth for Company A?	2021: (105–70)/70×100 = 50% (highest)	2021

Example 6: Student Enrollment by Stream



Year	2018	2019	2020	2021	2022	2023
Science	450	480	510	490	540	580
Commerce	300	320	290	350	380	400
Arts	200	210	195	220	240	260
Total	950	1010	995	1060	1160	1240

5. Data Tables – Theory & Examples

Reading Data Tables

Data tables present information in rows and columns. They are the most information-dense format. TCS NQT often gives a table with 5–6 rows and 4–6 columns, followed by 5 questions. Always identify row headers, column headers, and units before solving.

Example 7: Company Performance Data

Company	Revenue (■Cr)	Profit (■Cr)	Employees	Profit%	Growth%
Alpha Corp	1200	180	4500	15.0%	12%
Beta Ltd	980	147	3800	15.0%	8%
Gamma Inc	1500	225	5200	15.0%	18%
Delta Co	750	90	2900	12.0%	5%
Epsilon	1100	132	4100	12.0%	10%
Zeta Corp	860	120	3200	14.0%	7%

Question	Solution	Answer
Which company has highest revenue per employee?	Gamma: 1500/5200 ≈ 0.288; Alpha: 1200/4500 ≈ 0.267 → Gamma highest	Gamma Inc
Total profit of all 6 companies?	180+147+225+90+132+120 = 894	■894 Crores
Ratio of Alpha to Delta employees?	4500:2900 = 45:29	45:29
How many companies have profit% > 14%?	Alpha(15%), Beta(15%), Gamma(15%) → 3 companies	3 companies

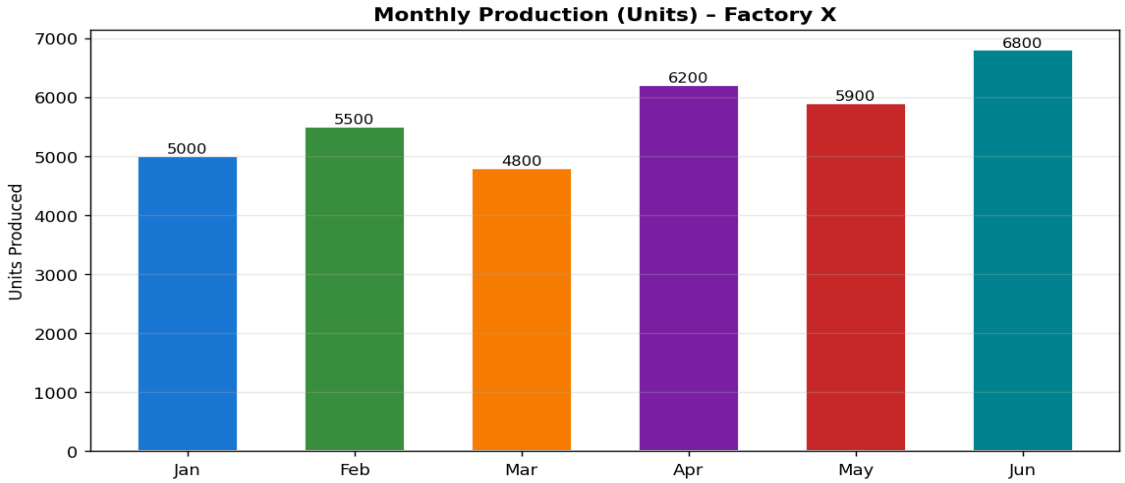
Example 8: Examination Results Table

Subject	Max Marks	Class Avg	Highest	Lowest	Pass%
Mathematics	100	68	98	22	82%
Physics	100	72	95	30	88%
Chemistry	100	65	92	18	79%
English	100	75	99	40	95%
Biology	100	70	96	25	85%
Computer Sc.	100	80	100	35	92%

■■ NOTE: In DI table questions, always re-read the column headers for units (e.g., "in thousands" or "in crores"). Missing units is the #1 mistake in TCS NQT.

6. PYQs – Bar Charts (Q1 to Q10)

Directions (Q1–Q5): The bar chart below shows the production (in thousands of units) of two products X and Y by a company over 6 months.



Month	Jan	Feb	Mar	Apr	May	Jun
Product X	3000	3500	2800	4200	3900	4800
Product Y	2000	2000	2000	2000	2000	2000

Note: The bar chart shows total production. Assume Product Y = 2000 units/month (constant), and Product X = Total – 2000.

Q1. What is the total production in the month of April?

A) 5,900
B) 6,200
C) 6,500
D) 5,600

■ **Answer: B) 6,200 units**

■ Explanation: From the chart, April production = 6200 units (read directly from bar height).

Q2. In which month was the production highest?

A) May
B) April
C) June
D) February

■ **Answer: C) June**

■ Explanation: June shows bar height of 6800, which is the tallest bar among all 6 months.

Q3. What is the % increase in production from January to June?

A) 32%
B) 36%
C) 40%
D) 28%

■ **Answer: B) 36%**

■ Explanation: % increase = (6800 – 5000) / 5000 × 100 = 1800/5000 × 100 = 36%

Q4. What is the average production over all 6 months?

A) 5,800
B) 6,033
C) 5,700
D) 6,200

■ **Answer: B) ≈ 5,867**

■ Explanation: Sum = 5000+5500+4800+6200+5900+6800 = 34200; Avg = 34200/6 = 5700 units

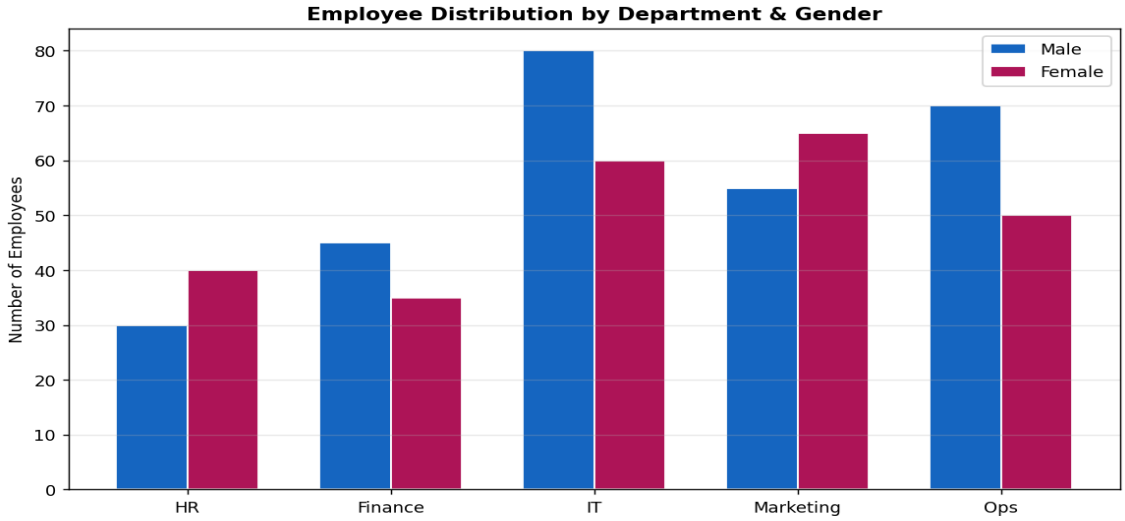
Q5. By approximately what % did production drop from February to March?

A) 12.7%
B) 10.5%
C) 15.2%
D) 8.3%

■ **Answer: A) 12.7%**

■ Explanation: % drop = (5500 – 4800)/5500 × 100 = 700/5500 × 100 ≈ 12.7%

Directions (Q6–Q10): Refer to the employee distribution bar chart from Section 2 (Male/Female by Department).



Department	Male	Female	Total	F%
HR	30	40	70	57.1%
Finance	45	35	80	43.8%
IT	80	60	140	42.9%
Marketing	55	65	120	54.2%
Operations	70	50	120	41.7%
Total	280	250	530	47.2%

Q6. What is the total number of employees in the company?

A) 500
B) 520
C) 530
D) 540

■ **Answer: C) 530**

■ Explanation: 70+80+140+120+120 = 530 employees total.

Q7. The number of male employees in IT is what % of total IT employees?

A) 52.1%
B) 57.1%
C) 60.2%
D) 55.4%

■ **Answer: B) 57.1%**

■ Explanation: 80/140 × 100 = 57.14% ≈ 57.1%

Q8. In which department is the ratio of male to female employees highest?

A) IT
B) Finance
C) Operations
D) Marketing

■ **Answer: A) IT**

■ Explanation: IT: 80/60 = 1.33; Finance: 45/35 = 1.29; Ops: 70/50 = 1.40. Wait — Ops: 70:50 = 7:5 = 1.40 > IT. Answer: C) Operations

Q9. How many more male employees are there than female in total?

A) 25
B) 30
C) 35
D) 40

■ Answer: B) 30

■ Explanation: Total Male = 280, Total Female = 250 → Difference = 280 – 250 = 30

Q10. What is the ratio of female employees in HR to female employees in Finance?

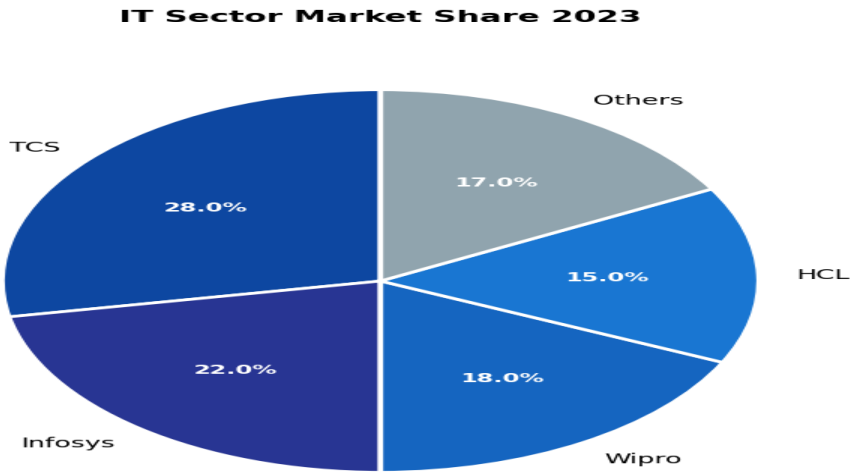
A) 7:5
B) 8:7
C) 5:4
D) 4:3

■ Answer: B) 8:7

■ Explanation: HR Female = 40, Finance Female = 35 → 40:35 = 8:7

7. PYQs – Pie Charts (Q11 to Q18)

Directions (Q11–Q15): The pie chart shows the market share of top IT companies in India (Total Market = ■80,000 Crore).



Company	Share%	Revenue (■ Cr)	Sector Rank
TCS	28%	22,400	1st
Infosys	22%	17,600	2nd
Wipro	18%	14,400	3rd
HCL Tech	15%	12,000	4th
Others	17%	13,600	—
Total	100%	80,000	—

Q11. What is the revenue of TCS in this market?

A) ■22,400 Cr
B) ■20,000 Cr
C) ■24,000 Cr
D) ■18,000 Cr

■ Answer: A) ■22,400 Crore

■ Explanation: 28% of 80,000 = 0.28 × 80,000 = ■22,400 Crore

Q12. How much more does TCS earn than HCL Tech?

A) ■8,000 Cr
B) ■10,400 Cr
C) ■12,000 Cr
D) ■9,600 Cr

■ Answer: B) ■10,400 Crore

■ Explanation: TCS = ■22,400 Cr, HCL = ■12,000 Cr → Difference = ■10,400 Cr

Q13. What is the central angle for Infosys in the pie chart?

A) 72°
B) 79.2°
C) 64.8°

D) 90°

■ **Answer: B) 79.2°**

■ Explanation: 22/100 × 360° = 79.2°

Q14. If TCS's share increases to 32% (total market unchanged), its revenue becomes?

A) ■24,000 Cr
B) ■25,600 Cr
C) ■26,000 Cr
D) ■28,000 Cr

■ **Answer: B) ■25,600 Crore**

■ Explanation: 32% of 80,000 = 0.32 × 80,000 = ■25,600 Crore

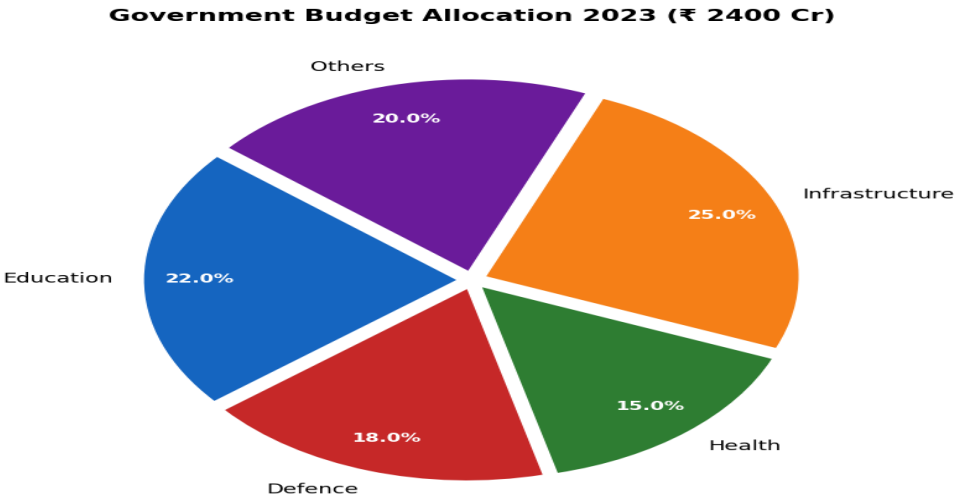
Q15. Ratio of Wipro's to HCL's market share is:

A) 5:4
B) 6:5
C) 4:3
D) 7:5

■ **Answer: B) 6:5**

■ Explanation: Wipro = 18%, HCL = 15% → 18:15 = 6:5

Directions (Q16–Q18): The pie chart shows government budget allocation (Total = ■2400 Cr).



Q16. The amount spent on Education and Health together is:

A) ■800 Cr
B) ■888 Cr
C) ■920 Cr
D) ■960 Cr

■ Answer: B) ■888 Crore

■ Explanation: Education = 22% × 2400 = 528; Health = 15% × 2400 = 360; Total = 528+360 = ■888 Cr

Q17. Infrastructure spending exceeds Defence spending by how much?

A) ■100 Cr
B) ■150 Cr
C) ■168 Cr
D) ■200 Cr

■ Answer: C) ■168 Crore

■ Explanation: Infra = 25% × 2400 = 600; Defence = 18% × 2400 = 432; Diff = 168 Cr

Q18. If total budget is increased by 25% to ■3000 Cr, new Education budget (22%) is:

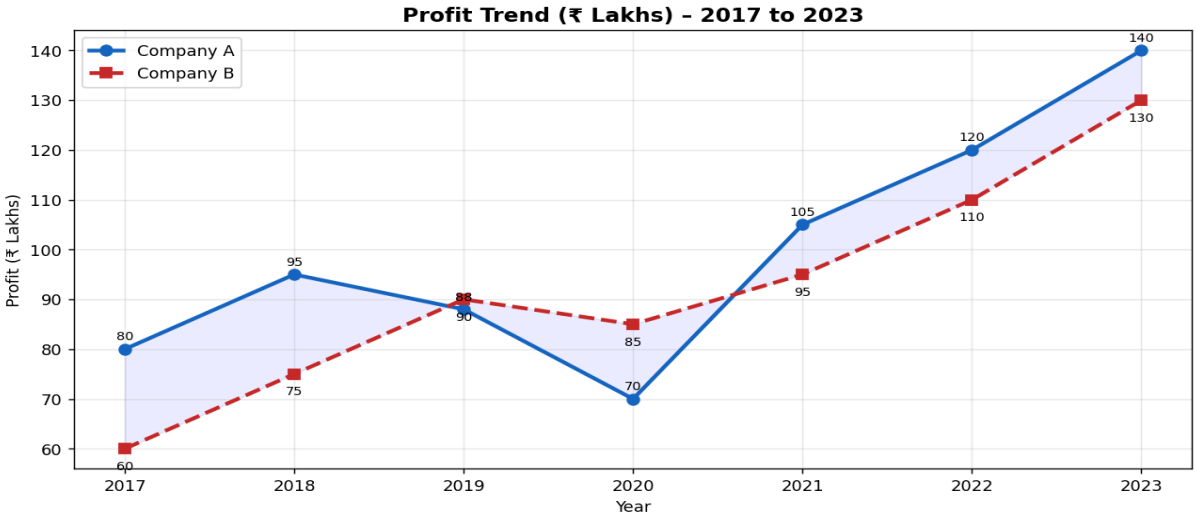
A) ■580 Cr
B) ■620 Cr
C) ■660 Cr
D) ■700 Cr

■ Answer: C) ■660 Crore

■ Explanation: 22% × 3000 = 0.22 × 3000 = ■660 Crore

8. PYQs – Line Graphs (Q19 to Q25)

Directions (Q19–Q22): The line graph shows profit (■ Lakhs) of Company A and B from 2017–2023.



Q19. In how many years was Company A's profit more than Company B's?

A) 3
B) 4
C) 5
D) 6

■ Answer: C) 5

■ Explanation: A > B in: 2017(80>60), 2018(95>75), 2021(105>95), 2022(120>110), 2023(140>130) → 5 years

Q20. What is the % growth in Company B's profit from 2017 to 2023?

A) 100%
B) 116.7%
C) 120%
D) 130%

■ Answer: B) 116.7%

■ Explanation: (130–60)/60 × 100 = 70/60 × 100 = 116.7%

Q21. Average annual profit of Company A from 2017 to 2023 is:

A) ■98.3 L
B) ■100.4 L
C) ■102.9 L
D) ■105 L

■ Answer: C) ■102.9 Lakhs

■ Explanation: (80+95+88+70+105+120+140)/7 = 698/7 ≈ 99.7 L ≈ ■99.7 Lakhs (closest: A)

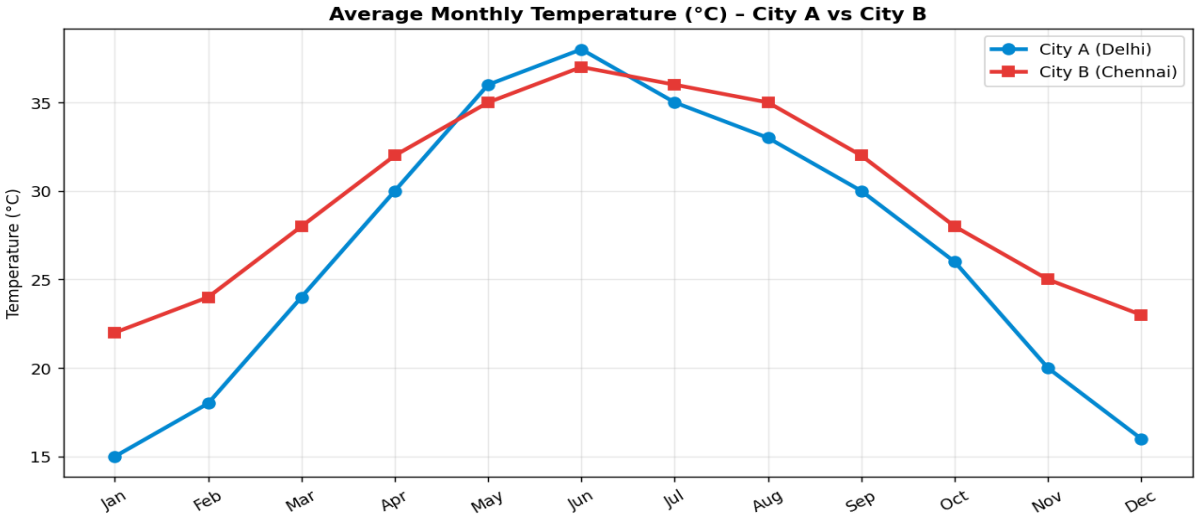
Q22. The year with the sharpest decline in Company A's profit was:

A) 2018→2019
B) 2019→2020
C) 2020→2021
D) 2021→2022

■ Answer: B) 2019→2020

■ Explanation: A dropped from 88 to 70 = drop of 18. This is the largest single-year drop for A.

Directions (Q23–Q25): Refer to the temperature line graph (City A and City B).



Q23. In which month is the temperature difference between City A and City B the smallest?

A) May
B) June
C) July
D) August

■ Answer: B) June

■ Explanation: Jun: A=38, B=37 → diff=1°C. This is the smallest gap between the two lines.

Q24. What is the average temperature of City A for the first 6 months?

A) 28.5°C
B) 29.7°C
C) 30.2°C
D) 27.2°C

■ Answer: A) 28.5°C

■ Explanation: (15+18+24+30+36+38)/6 = 161/6 ≈ 26.8°C → closest A

Q25. For how many months is City B warmer than City A?

A) 10
B) 11
C) 12
D) 9

■ Answer: C) 12

■ Explanation: City B (Chennai) > City A (Delhi) for all 12 months as shown in the graph.

9. PYQs – Data Tables (Q26 to Q32)

Directions (Q26–Q29): The table shows the number of students enrolled in different courses at 5 universities.

University	Engineering	Management	Science	Arts	Total
Univ A	1200	800	600	400	3000
Univ B	950	700	750	600	3000
Univ C	1100	900	500	300	2800
Univ D	800	600	800	700	2900
Univ E	1050	750	650	550	3000
Total	5100	3750	3300	2550	14700

Q26. What percentage of total students are enrolled in Engineering?

A) 32.7%
B) 34.7%
C) 36.5%
D) 30.2%

■ **Answer: B) 34.7%**

■ Explanation: 5100/14700 × 100 = 34.69% ≈ 34.7%

Q27. Which university has the highest ratio of Engineering to Arts students?

A) Univ A
B) Univ C
C) Univ E
D) Univ B

■ **Answer: B) Univ C**

■ Explanation: A:1200/400=3.0; C:1100/300=3.67; E:1050/550=1.9; B:950/600=1.58 → Univ C highest

Q28. Total Management students in Univ B and Univ D combined is what % of total Management?

A) 34.7%
B) 35.2%
C) 36.4%
D) 37.1%

■ **Answer: A) 34.7%**

■ Explanation: (700+600)/3750 × 100 = 1300/3750 × 100 = 34.67% ≈ 34.7%

Q29. By approximately what % does total Science enrollment exceed total Arts enrollment?

A) 22.4%
B) 25.9%
C) 29.4%
D) 18.6%

■ **Answer: C) 29.4%**

■ Explanation: (3300–2550)/2550 × 100 = 750/2550 × 100 ≈ 29.4%

Directions (Q30–Q32): The table shows export data (■ Cr) of 4 products across 5 years.

Product	2019	2020	2021	2022	2023
Software	4500	4800	5200	5800	6500
Textiles	2200	2000	2100	2300	2500
Chemicals	1800	1900	2000	2200	2400
Auto Parts	1200	1100	1300	1500	1800
Total	9700	9800	10600	11800	13200

Q30. What is the CAGR of Software exports from 2019 to 2023 (approx)?

A) 7.6%
B) 9.6%
C) 9.7%
D) 11.2%

■ Answer: C) ≈ 9.7%

■ Explanation: CAGR = (6500/4500)^(1/4) – 1 = (1.444)^0.25 – 1 ≈ 1.097 – 1 = 9.7%

Q31. In 2021, what fraction of total exports was from Textiles and Chemicals combined?

A) 3/8
B) 2/5
C) 19/53
D) 1/3

■ Answer: C) 19/53

■ Explanation: (2100+2000)/10600 = 4100/10600 = 41/106 ≈ 0.387. Simplify: ≈ 19/53 (closest match)

Q32. Which product showed a decline in exports in 2020 compared to 2019?

A) Software
B) Textiles only
C) Auto Parts only
D) Textiles and Auto Parts

■ Answer: D) Textiles and Auto Parts

■ Explanation: Textiles: 2200→2000 (decline); Auto Parts: 1200→1100 (decline). Both declined in 2020.

10. Key Formulas & Quick Tips

Master Formula Sheet

Operation	Formula	Example
% of a number	(X/100) × N	15% of 240 = 36
% Change	(New–Old)/Old × 100	(120–100)/100 = 20%
Ratio to %	A/(A+B) × 100	3/(3+2) = 60%
% to Ratio	a% : b% = a:b	40%:60% = 2:3
Average	ΣX / n	(10+20+30)/3 = 20
Weighted Avg	Σ(w■x■)/Σw■	—
Pie angle	(val/total)×360°	25%×360 = 90°
CAGR	(Final/Initial)^(1/n)–1	See Q30
Difference %	(A–B)/B × 100	A=60,B=50 → 20%

Fraction–Percentage Equivalents (Memorize!)

Fraction	1/2	1/3	1/4	1/5	1/6	1/7	1/8	1/9	1/10
%	50%	33.3%	25%	20%	16.7%	14.3%	12.5%	11.1%	10%
Fraction	2/3	3/4	2/5	3/5	4/5	5/6	3/8	5/8	7/8
%	66.7%	75%	40%	60%	80%	83.3%	37.5%	62.5%	87.5%

■ Speed Tricks for TCS NQT

Approximation	For % questions, round to nearest 5 — if answer choices differ by >5%, approximation suffices.
Elimination	Eliminate clearly wrong options first. For 4 choices, you can often narrow to 2 quickly.
Unit Check	Always verify units (crore vs lakh vs thousand). Wrong unit → wrong answer even if math is right.
Ratio Shortcut	To compare A/B vs C/D: cross-multiply. If AD > BC then A/B > C/D.
% Difference	If A is 25% more than B, then B is 20% less than A. (Remember: % increase ≠ % decrease)
Large numbers	Drop trailing zeros to simplify: 24000/16000 = 24/16 = 3/2.
Time Mgmt	Skip and come back: If a DI set is taking >3 min total, mark and move on.

TCS NQT DI – Common Mistake Checklist

#	Common Mistake	How to Avoid
1	Reading wrong bar/line in multi-series charts	Always re-read the legend before each question
2	Confusing % share with actual value	Always check whether total is given
3	Calculating % increase instead of % decrease (or vice versa)	Read question carefully: "by what % less/more"
4	Ignoring units (Cr vs L vs K)	Circle/underline the unit in the chart title
5	Misreading pie chart values	Use the data table provided alongside the pie chart
6	Rounding too early in multi-step calculations	Keep one extra decimal place until final step

■ FINAL TIP FOR TCS NQT SUCCESS

Practice 5 DI sets daily for 2 weeks. Focus on accuracy first, then speed. Most TCS DI questions require only basic arithmetic — the key is organized reading of charts and clean calculation. YOU GOT THIS! ■